Task Investment to First Place: Aggregation Rules and Score Ceilings

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Abstract

Multi-task leaderboards can reward improvements concentrated on selected tasks. We define conditional investment as the minimum number of target task scores that must improve to reach tied first under a specified score envelope, with rivals fixed and no transfer across tasks. Exact optimization on HELM MMLU (79 entries, 57 subjects) and the Open LLM Leaderboard (4,575 entries, six aggregates) compares mean, median, geometric mean and Borda. With selected scores raised to perfection, median costs are respectively 16, 21, 16 and 46 subjects on HELM, and 2, 3, 3 and 6 aggregates on Open LLM. Yet replacing perfection with per-task observed frontiers reverses the mean–median ordering by average minimum cost on both boards; partial uplift can prevent victory altogether. Random-task baselines and task, representation and competitor-pool checks delimit these findings. Aggregation affects the required breadth of score improvement, but its comparison depends on attainable-score assumptions. These are reproducible score-table counterfactuals, not measurements of training expense or contamination.

1 Introduction

A multi-task leaderboard turns a vector of task scores into one ordering. Developers may respond to that ordering by concentrating data collection or adaptation on selected tasks. Even without estimating how training changes scores, we can ask a precise preliminary question: how many task coordinates would have to improve for a particular entry to reach first place, and how does the aggregation rule change that number?

We study this question as a conditional score intervention on two public snapshots: HELM MMLU and the Open LLM Leaderboard. Investment is the number of directly affected tasks. A selected coordinate is raised to a declared endpoint; other coordinates and all rivals' raw scores stay fixed. This measures the breadth of task-specific intervention, a useful quantity when access to additional task-related data is itself constrained. It does not equate task breadth with training examples, computation or money. A six-task aggregate and a subject-level task also represent different intervention units.

The central finding is that the attainable-score envelope can reverse a comparison between rules. With perfect selected scores, mean aggregation requires fewer affected tasks on average than median aggregation in both snapshots. Replacing perfect scores by the observed per-task frontier reverses this average-cost ordering in both. Under partial uplift, some targets cannot win even when every task is available. Thus a resistance claim must specify what the intervention can achieve, as well as the aggregation rule.

Our contributions are a precise fixed-table investment definition, exact optimization with retained witnesses, and a reproducible comparison of arithmetic mean, median, geometric mean and Borda on pinned public matrices. We report full budget–success distributions, paired ceiling comparisons,

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

random-task baselines, and task/representation/pool sensitivities. These results describe conditional first-place accessibility; they do not identify contamination or establish a universally most resistant aggregation rule.

2 Related work

MMLU evaluates knowledge across subject tasks [Hendrycks et al., 2021]; HELM motivates transparent, explicitly scoped evaluation [Liang et al., 2023]. BBH illustrates why a benchmark aggregate is different from an individual subtask [Suzgun et al., 2022]. The Benchmark Lottery already shows that task selection and aggregation can change model comparisons [Dehghani et al., 2021]. Our object is instead a minimum number of target coordinates that must improve to attain first, conditional on the allowed endpoints.

Social-choice evaluation treats tasks as voters and models as alternatives [Lanctot et al., 2025]. Our Borda intervention raises one alternative in each selected task ordering. It is related to restricted shift bribery [Faliszewski and Rothe, 2016], but uses endpoints induced by score changes, rather than unrestricted ballot replacement or arbitrary shift prices. Ordinal invariance, clone consistency and minimum intervention cost are different properties.

Contamination experiments show that benchmark-related training can inflate measured performance, sometimes with losses on other tasks [Zhou et al., 2023, Yang et al., 2023]. They motivate the question without validating independent score saturation for every listed model. Leaderboard reporting practices [Singh et al., 2025] and strategic post-training equilibria [Chen et al., 2026] concern additional mechanisms. We neither infer historical misconduct nor model simultaneous developer responses. Finally, uncertainty in aggregate benchmark estimates [Agarwal et al., 2021] should not be confused with optimization uncertainty: our optima condition on published scores, while subject-subset ranges describe perturbations of those tasks.

3 Conditional task investment

Let $X \in [0,1]^{N\times T}$ be a complete, higher-is-better score matrix, and let i be the target entry. For an affected set $S \subseteq \{1,\dots,T\}$, set

$$X_{ij}^{S} = \begin{cases} y_{ij} & j \in S, \\ X_{ij} & j \notin S, \end{cases} \qquad y_{ij} = X_{ij} + \alpha(c_j - X_{ij})_+, \tag{1}$$

where $(u)_+ = \max(u,0)$ and all other rows remain fixed. We use $c_j = 1$ (perfect ceiling) or $c_j = \max_\ell X_{\ell j}$ (observed frontier), with $\alpha \in \{1/4, 1/2, 1\}$. These are declared hypothetical envelopes, not estimates of attainable training gains.

For rule r, write $A_r(X)_i$ for the target's aggregate score. The minimum investment is

$$K_i(r;\alpha,c) = \min_{S} \left\{ |S| : A_r(X^S)_i \geq \max_{\ell\neq i} A_r(X^S)_\ell \right\}. \tag{2}$$

An empty feasible family gives $K_i = \infty$; initial winners have $K_i = 0$. Our primary criterion is tied first. Strict first is a sensitivity. Cardinal comparisons use tolerance 10^{-11} on the unit scale; strict first requires a margin exceeding that tolerance. Borda uses exact doubled integer points and one additional doubled point for strict first.

Rules. We compare $T^{-1}\sum_j X_{ij}$, the conventional median (averaging the middle two values when T is even), and $(\prod_j X_{ij})^{1/T}$. A zero coordinate makes the geometric mean zero; no pseudocount is added. Borda averages within-task pairwise points: one point for a lower-scoring rival, half a point for a tie, and zero for a higher rival. Equivalently it sums ascending average ranks minus one. Its normalized score divides by $T(N-1)$; this does not change winners. Within-task ties use equality of parsed published scores. Borda recomputes *every* entry's rank points after intervention: fixed raw rival scores do not imply fixed rival Borda totals.

Why task count? The unit cost operationalizes breadth of direct task-specific intervention without inventing a training-response function from evaluation tables. It assumes equal prices for affected

coordinates, independent uplifts, no collateral losses, and unilateral action. These assumptions isolate an aggregation mechanism and make an auditable finite problem. They are restrictive: one training dataset may transfer across tasks, subjects may differ greatly in difficulty, and an aggregate may contain many subtasks. Consequently K/T is descriptive coverage within a declared partition, not a common physical currency across boards.

Envelope interpretation. All four rules weakly favor raising the target coordinate. Enlarging the permitted independent endpoints cannot increase K: any successful set under the smaller envelope remains successful under the larger one. Perfect saturation therefore gives an optimistic affected-coordinate bound inside this model when real per-coordinate gains cannot exceed that ceiling. This statement does not lower-bound the number of real training datasets when data transfer across coordinates. Nor does it validate either envelope as achievable.

3.1 Exact computation

For the mean, sort the gains $y_{ij} - X_{ij}$ and take the shortest prefix crossing the best rival's aggregate. For the geometric mean use log gains, repairing zero coordinates before a positive aggregate can be reached. If the rival threshold is zero, a zero-cost tie is already sufficient. For odd T, a median reaches rival threshold h exactly when at least $(T+1)/2$ coordinates reach h. Count the deficit among already qualifying coordinates and check whether enough endpoints can cross h. The six-task board enumerates all subsets, covering its even median without approximation.

For Borda, let P_ℓ be entry ℓ's initial doubled point total. Let $D_{\ell j}$ be the increase in target pairwise points against rival ℓ if task j is selected, and $G_j = \sum_{\ell \neq i} D_{\ell j}$. Selecting task j raises the target by G_j and lowers rival ℓ by $D_{\ell j}$. The exact binary covering problem is

$$\min_{z \in \{0,1\}^T} \sum_j z_j, \qquad \sum_j (G_j + D_{\ell j}) z_j \geq P_\ell - P_i \quad (\ell \neq i). \tag{3}$$

A rival loses at most $2T$ doubled points, so rivals initially more than $2T$ below the highest rival cannot bind. We remove these redundant constraints, solve the HELM problem with SciPy/HiGHS mixed-integer programming at zero relative gap, and enumerate all subsets for Open LLM. Returned witnesses are checked against all rivals. Independent small-instance exhaustive tests exercise the analytic and integer-program paths, including zeros, ties, strictness and partial/frontier uplift. Appendix A gives further details.

4 Public snapshots and study design

HELM MMLU. We use release v1.13.0 (release date 10 January 2025): 79 complete entries across 57 MMLU subjects. All repeated subject-table representations agree on their score/count pairs; they are collapsed rather than counted twice. No score is imputed. Our main arithmetic mean weights subjects equally. A count-weighted sensitivity uses the published evaluation-item counts, which vary across subjects.

Open LLM. We pin the public `contents` snapshot at commit `9c09a7cae43334062a82cb164f2ef255013dafa2` (20 March 2025). It contains 4,576 complete rows; excluding its one flagged row gives 4,575 primary entries. Tasks are the six published aggregates IFEval, BBH, MATH Lvl 5, GPQA, MUSR and MMLU-PRO. We divide official normalized percentage columns by 100; their mean exactly reproduces the snapshot's reported Average. Published Raw columns are a sensitivity, not assumed to be an order-preserving transform of the aggregate columns. Entries can share model families, fullnames or score vectors and are not independent training replications.

Both snapshots were acquired on 3 November 2026. Source URLs, timestamps and SHA-256 values are in `code/data/manifest.json`; processed matrices and audits are included. We use existing score summaries only, with no new training or inference. Public score-data licenses are not separately declared in the supplied sources; we do not infer a data license from code licenses or model metadata. Reproduction retrieves raw files from the original sources.

Table 1: Perfect-score saturation: mean and median minimum cost, interquartile interval, and mean task fraction. All 79 HELM and 4,575 Open LLM entries are included, including initial winners ($K = 0$). Quartiles interpolate the empirical distribution; fractional quartiles are not attainable task counts. IQRs describe heterogeneity across snapshot entries, not estimation uncertainty.

Board	Rule	Mean K	Median K	IQR	Mean K/T
HELM MMLU	Mean	16.70	16	[10, 22.5]	29.3%
HELM MMLU	Median	19.32	21	[13.5, 27]	33.9%
HELM MMLU	Geometric	16.92	16	[10, 22.5]	29.7%
HELM MMLU	Borda	42.23	46	[38, 51.5]	74.1%
Open LLM	Mean	2.36	2	[2, 3]	39.4%
Open LLM	Median	2.85	3	[3, 3]	47.5%
Open LLM	Geometric	2.98	3	[2, 4]	49.7%
Open LLM	Borda	5.51	6	[5, 6]	91.8%

Comparisons and uncertainty. Primary analyses evaluate every entry under every rule, $\alpha = 1$, $c = 1$, tied first. We report the empirical budget–success function

$$F_r(b) = N^{-1} \sum_i \mathbf{1}\{K_i(r) \leq b\}, \tag{4}$$

so infeasible targets remain in the denominator. Rule differences pair the same targets; all summaries include initial winners unless stated otherwise. We vary ceiling, uplift and strictness; compare optimal task selection with uniform random task order; perturb the HELM task set; and test raw scores, entry deduplication, the flagged row and item weighting. All optimization rows have certified-optimal or infeasible status; none are reported from an unresolved solve.

Fixed-snapshot optima have no Monte Carlo optimization error. Across-entry quartiles describe heterogeneity, not sampling uncertainty about a model population. To expose sensitivity to task selection, we re-optimize 200 uniform subsets of 45 distinct HELM subjects (seed 449027). These central empirical ranges concern omission perturbations of observed subjects, not future-task confidence intervals or uncertainty in true accuracies. For a random-order baseline, Open LLM enumerates all subsets exactly; HELM uses 2,000 orders per target shared across rules (seed 519733). Its per-target Wilson intervals quantify Monte Carlo error for success probabilities only. Marginal Monte Carlo errors for the across-entry expected costs are computed separately from stopping-cost variances (Appendix B); entries remain fixed.

5 Results

5.1 How much task coverage under perfect scores?

Table 1 answers the investment question under the primary envelope. Median minimum counts across HELM entries are 16/57 for mean, 21/57 for median, 16/57 for geometric mean, and 46/57 for Borda. Open LLM counts are respectively 2/6, 3/6, 3/6 and 6/6. The full distributions in Figure 1 retain the variation that a single typical count hides. These are different task partitions, so their fractions do not compare actual training budgets.

The mean–median difference is also visible within targets: mean costs less for 64 of 79 HELM entries, equally much for three and more for 12. On Open LLM the counts are 2,165 less, 2,409 equal and one more. Borda has higher average cost than the three cardinal rules on both primary snapshots; this is an empirical property of these populations and endpoints, not a consequence of ordinal invariance or a universal protection guarantee.

Starting position matters even within the same envelope. Using a common arithmetic-mean ranking to define the top decile, its eight HELM entries have average costs 2.50, 2.63, 2.50 and 10.88 under mean, median, geometric mean and Borda. The corresponding 458 Open LLM entries have averages 1.11, 2.14, 1.37 and 3.47. These fixed-reference strata hold target membership constant across rules; they describe position in each snapshot rather than matched model quality.

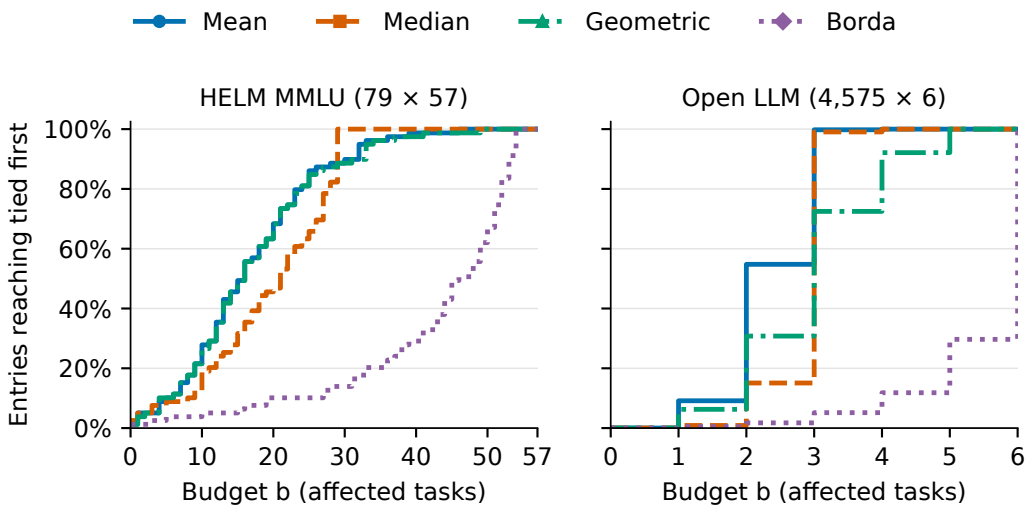


Figure 1: Minimum number of affected tasks needed to reach tied first under full perfect-score saturation. Each step is the fraction of all entries with exact cost $K \leq b$. Panel headings show entries $\times$ tasks. Opponent scores are fixed; unselected tasks do not change and there is no cross-task transfer. HELM tasks are 57 subjects, while Open LLM tasks are six published aggregates. The counts measure score interventions, not observed training investment.

5.2 A ceiling change reverses the mean–median ordering

Under perfect saturation, average K is 16.70 for mean versus 19.32 for median on HELM, and 2.36 versus 2.85 on Open LLM. With selected scores raised only to observed frontiers, these become 34.09 versus 19.32 and 3.88 versus 3.24, respectively (Figure 2). All targets are feasible under both full-uplift envelopes. Thus the average-cost reversal compares identical populations, without conditioning on different successful subsets. Paired counts appear in Appendix B.

Why can the difference be so large? A perfect selected score can provide substantial surplus to a cardinal average, offsetting untouched weak tasks. A frontier caps this surplus. The odd-task median instead needs enough coordinates crossing the best rival median; full HELM frontiers still allow those crossings, leaving its minimum cost unchanged. This explanation follows the optimization structure, not a fitted training-response model.

Partial uplift makes attainability decisive. At half headroom toward perfect scores, only 40/79 HELM targets can win under mean, 41/79 under median, 42/79 under geometric mean and 32/79 under Borda. Under half headroom toward observed frontiers the counts fall to 4/79, 7/79, 4/79 and 6/79. For Open LLM, half-perfect feasibility is 4,423/4,575, 3,428/4,575, 4,575/4,575 and 1,973/4,575; half-frontier feasibility is 160/4,575, 463/4,575, 164/4,575 and 632/4,575. We report feasibility rather than compare conditional cost means on different sets (Appendix B).

5.3 Strategic selection versus a random task order

Under perfect saturation, choosing the task set optimally reduces average task count relative to a uniform random order (Figure 3). On HELM, the optimal versus random expected counts are 16.70 versus 27.23 for mean, 19.32 versus 21.90 for median, 16.92 versus 29.94 for geometric mean, and 42.23 versus 47.89 for Borda. For Open LLM they are 2.36 versus 2.74, 2.85 versus 3.02, 2.98 versus 3.95 and 5.51 versus 5.83. Random order is a reference for the value of task selection, not a behavioral model of developers. HELM random expectations are Monte Carlo estimates; the optima and Open LLM expectations are exact finite-table quantities. HELM standard errors, in the stated rule order, are 0.0052, 0.0037, 0.0063 and 0.0034 tasks; they concern permutation simulation only.

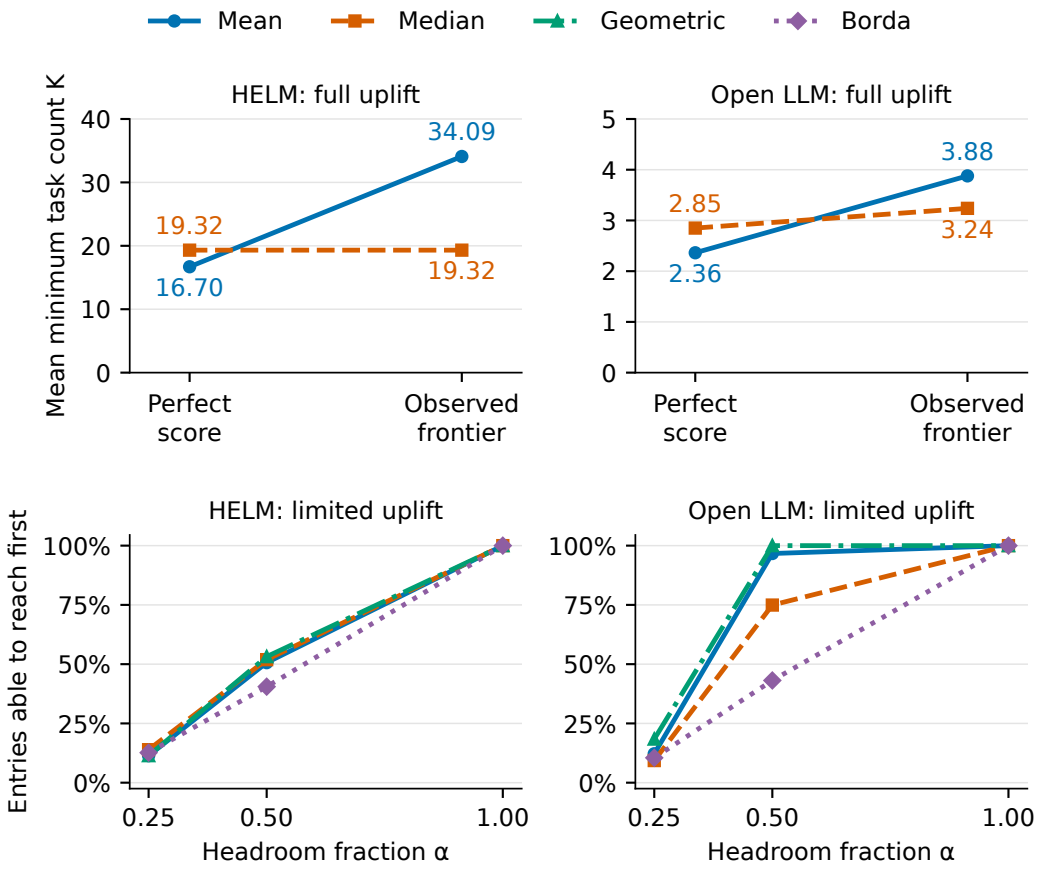


Figure 2: Attainable scores change both cost ordering and feasibility. Top: mean minimum task count for mean and median aggregation when selected scores reach either 1 or the per-task observed frontier. All targets are feasible in these full-uplift settings, so both points use identical populations. Bottom: fraction of all targets able to win, even with all tasks available, for three fractions of remaining headroom toward 1. Lines join evaluated settings only. All panels use tied first, fixed opponents and no cross-task transfer. These hypothetical ceilings are assumptions, not estimates of achievable training improvements.

5.4 Scope checks

Across 200 distinct-subject omission perturbations, the central 95% range of paired mean-minus-median average costs is $[-3.49, -1.25]$ tasks, and mean-minus-Borda is $[-20.75, -19.27]$ (45 subjects retained). This supports the primary ordering within the declared perturbation distribution, without resolving unobserved task or score uncertainty. Appendix B shows the coverage ranges.

Changing the score representation or competitor pool can matter. Raw versus normalized Open LLM columns change K for 268 mean, 458 median, 1,190 geometric and 64 Borda targets. Several published aggregate columns change within-task ranks, so this is not a test of Borda invariance to strictly increasing transforms. Removing repeated score vectors changes Borda cost for 11 of 4,521 retained entries; removing repeated fullnames changes it for seven of 4,496. Cardinal costs are unchanged for retained entries in these two removal checks. These small specific effects do not establish clone consistency. Reinstating the flagged entry changes no shared target cost. Weighting HELM's mean by evaluation-item counts changes its perfect-ceiling average from 16.70 to 11.13 while retaining task-count cost, underscoring that task weights and investment prices are separate choices. Strict-first configurations and all sensitivity rows remain available in the code package.

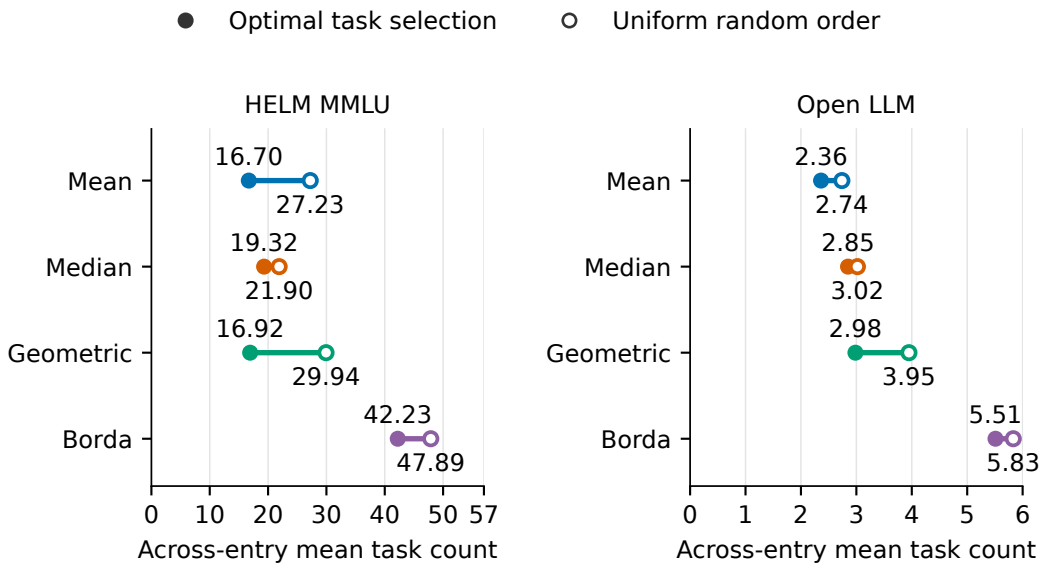


Figure 3: Strategic task selection versus a uniform random order under perfect-score saturation. Solid dots show across-entry mean exact optimal K; open dots show the across-entry average expected stopping cost for random task order. Open LLM expectations enumerate all subsets exactly. HELM expectations are Monte Carlo estimates from 2,000 shared-across-rule orders per target (seed 519733). Appendix Table 5 reports marginal Monte Carlo precision of the HELM means. All entries, including initial winners, are included.

6 Discussion and limitations

The answer depends jointly on the rule, starting scores, competitor pool, task partition and intervention envelope. In the perfect-score scenario, the cardinal rules permit first place with fewer affected tasks on average than Borda on these snapshots. The observed-frontier scenario overturns the mean–median average-cost comparison; partial uplift often prevents any successful intervention. A single saturation result is therefore insufficient to recommend a universally resistant aggregation rule.

The model isolates a useful structural question, but the investment unit is deliberately narrow. Equal task prices ignore unequal data requirements and score difficulty. Uplifts are not learned from training experiments; neither perfect scores nor observed rival frontiers are known attainable for a given target. Transfer can make direct coordinate count overstate the number of training datasets needed, while interference can make independent uplift optimistic. Published scores contain evaluation noise that these summary tables cannot fully quantify. Our exactness claims concern the optimization conditional on those values, not the true capabilities of the models.

The two boards also differ in granularity, model populations and score definitions. Related and merged model entries are not independent experimental units. Geometric mean depends on score origin and zeros; Borda discards cardinal margins and depends on the competitor population. The item-weighted and raw-score checks demonstrate sensitivity to particular conventions rather than identify a uniquely correct normalization. Task-omission ranges do not warrant generalization to future benchmarks, and our four rules do not exhaust aggregation choices. Finally, fixed rivals exclude strategic response and equilibrium effects.

An evaluation designer can use the released budget–success curves as a conditional audit: disclose task units and score envelopes, expose unreachable targets, and check whether a conclusion survives plausible changes of assumptions. Translating such an audit into training-data or compute costs requires an independently measured, target-specific response map, including transfer and collateral losses. Low conditional cost is not evidence that any named model was contaminated.

7 Conclusion

We define investment as the minimum number of directly improved task scores needed to reach first place and compute it exactly on two public multi-task leaderboard snapshots. Perfect-score saturation yields median costs of 16, 21, 16 and 46 subjects on HELM and 2, 3, 3 and 6 aggregates on Open LLM for mean, median, geometric mean and Borda. Crucially, restricting selected scores to observed frontiers reverses the mean–median average-cost ordering in both boards, while partial uplift can make victory infeasible. The resulting conclusion is conditional: aggregation affects task-breadth investment, but its comparison cannot be separated from assumptions about attainable scores.

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A Exactness, conventions and mathematical scope

Separable cardinal rules. For the arithmetic mean, the best rival aggregate is fixed, so success requires selected gains to cross a fixed sum gap. Among all sets of a given size, the largest gains maximize that sum; the first successful sorted prefix is optimal. The same argument applies to positive geometric scores in log space. If the target contains zeros and a positive rival threshold must be met, all repairable zeros must be repaired; an unrepairable zero makes success impossible. If the best rival geometric mean is zero, the tied-first cost is zero. For an odd median, let $m = (T + 1)/2$. At least m coordinates must reach the rival threshold; each eligible affected coordinate contributes one to this count. The cost is the deficit if there are enough eligible coordinates, and infinity otherwise. These arguments extend to the declared partial endpoints; they do not assume that training realizes those endpoints.

Borda constraints. Doubled pairwise points remove fractional ties. A target's upward movement on task j increases its score against rival ℓ by $D_{\ell j} \in \{0, 1, 2\}$ and decreases that rival's score by the same amount. Other rivals' mutual pairwise scores stay unchanged. Summing gives Equation (3). For pruning, choose an initially highest-scoring rival h. After any intervention it retains at least $P_h - 2T$ points. A rival initially below that value can never overtake h since it only loses points. Its constraint is redundant. The implementation retains equality-boundary rivals and checks the solution against the full opponent set.

Numerical and validation scope. Cardinal success is evaluated at tolerance 10^{-11}; these are tolerance-defined optima, rather than symbolic rational-accuracy claims. Borda uses integer constraints. HELM uses analytic cardinal solvers and certified mixed-integer Borda solutions; Open LLM enumerates its task subsets for every rule. The independent exhaustive test suite includes odd and even medians, zeros/ties, frontier ceilings, fractional uplift and strict first, including cases large enough to bypass the six-task shortcut. Full validation checks 296,064 primary/sensitivity rows and completion of 63,200 subject-subset optimizations. All full rows have optimal or infeasible status; there are no bounded or unresolved rows. Per-target selected tasks, lower bounds, statuses and solver time are retained. Synthetic matrices are used only to verify algorithms, not as empirical leaderboard evidence.

No universal mean–median ordering. The following are algebraic boundary examples, not measured leaderboard results. With three tasks and a perfect ceiling, competitor $(.9, .9, .1)$ and target $(.6, .6, .6)$ give costs one for mean and two for median: the target needs two coordinates above .9 to reach the rival median, but one perfect coordinate suffices for its mean. Conversely, competitor $(.95, .95, .95)$ and target $(0, 0, 0)$ give costs three for mean and two for median. Thus neither rule always requires greater task count, even with the same intervention envelope and an initially losing target under both. The observed empirical reversal strengthens the practical reason to state the ceiling, rather than deriving a universal ordering from either snapshot.

B Additional sensitivity evidence

Table 2 retains feasibility denominators for both half-uplift ceilings. Table 3 shows that the full-uplift reversal occurs on identical target populations: on HELM, mean costs less than median for 64 targets under perfect scores and none under frontiers; on Open LLM, those counts are 2,165 and ten. This is a statement about these targets and the declared statistic, not a universal pairwise dominance assertion. Table 4 gives the retained-target comparisons for score representation and competitor-pool checks. Figure 4 displays the subject-omission perturbations; independent model-sampling error bars would be inappropriate for these census summaries.

Monte Carlo precision of the HELM expected costs. For each fixed target i and rule r, let L_{irj} be the stopping count from permutation j, and let s_{ir}^2 be its sample variance with denominator $R - 1$. Separate target arrays use independent permutation draws, while rules share the same orders within a target. For $M = 79$ fixed entries and $R = 2{,}000$ orders each, the estimate and marginal variance estimate are

$$\widehat{\theta}_r = \frac{1}{M} \sum_i \frac{1}{R} \sum_j L_{irj}, \qquad \widehat{\mathrm{Var}}(\widehat{\theta}_r) = \frac{1}{M^2} \sum_i \frac{s_{ir}^2}{R}. \tag{5}$$

Table 2: Full-uplift average cost under perfect and observed-frontier ceilings (all entries feasible), and feasible/total entries at half remaining-headroom uplift under each ceiling. Feasible counts use every target as the denominator. Conditional costs under partial uplift are omitted to avoid comparisons between different feasible populations.

Board	Rule	Mean K perfect	Mean K frontier	Feasible half, perfect	Feasible half, frontier
HELM MMLU	Mean	16.70	34.09	40/79	4/79
HELM MMLU	Median	19.32	19.32	41/79	7/79
HELM MMLU	Geometric	16.92	33.28	42/79	4/79
HELM MMLU	Borda	42.23	43.51	32/79	6/79
Open LLM	Mean	2.36	3.88	4,423/4,575	160/4,575
Open LLM	Median	2.85	3.24	3,428/4,575	463/4,575
Open LLM	Geometric	2.98	4.65	4,575/4,575	164/4,575
Open LLM	Borda	5.51	5.51	1,973/4,575	632/4,575

Table 3: Paired mean versus median costs for the same targets under each full-uplift ceiling. The last three columns count targets for which mean costs less, equally much, or more than median. Here $\Delta K = K_{\mathrm{mean}} - K_{\mathrm{median}}$; a positive mean paired difference indicates larger mean-rule cost. All targets are feasible.

Board	Ceiling	Mean ΔK	Lower	Equal	Higher
HELM MMLU	Perfect	-2.62	64	3	12
HELM MMLU	Frontier	+14.77	0	3	76
Open LLM	Perfect	-0.49	2165	2409	1
Open LLM	Frontier	+0.64	10	1711	2854

Already-winning targets contribute zero mean and variance. Approximate marginal 95% intervals use $\widehat{\theta}_r \pm 1.959964\,\widehat{\mathrm{SE}}_r$. Table 5 reports the four intervals. These concern simulation precision for these fixed scores and targets; they neither describe a sampled model population nor provide simultaneous 95% coverage. No interval for a difference between rules is reported, since that calculation would require the shared-order covariance. Replaying seed 519733 preserves every original baseline column; the precision extension retains all 316 target/rule variances separately. An independent exact median-order-statistic check gives expected cost 21.90012 and Monte Carlo SE 0.00376 at this R, consistent with the estimated precision. Source data and that check are in `code/notes/RANDOM_BASELINE_MC_ERROR.md` and its linked result files. Open LLM expectations and optimized costs have no Monte Carlo error.

Table 4: Open LLM representation and competitor-pool checks, paired on retained entries. ΔK is variant minus primary cost under perfect-score saturation. Raw versus normalized published aggregates need not preserve within-task ranks, so Borda changes here do not test invariance to strictly increasing per-task transforms. Unique scores and unique fullnames retain one entry per score vector and fullname, respectively. These deduplications and inclusion of the flagged entry test only these specific pool changes.

Variant	Rule	Paired n	Changed K	Mean ΔK	Max $\lvert\Delta K\rvert$
Raw scores	Mean	4,575	268	-0.0144	1
Raw scores	Median	4,575	458	+0.0835	2
Raw scores	Geometric	4,575	1,190	-0.2299	1
Raw scores	Borda	4,575	64	-0.0039	1
Include flagged	Mean	4,575	0	0	0
Include flagged	Median	4,575	0	0	0
Include flagged	Geometric	4,575	0	0	0
Include flagged	Borda	4,575	0	0	0
Unique scores	Mean	4,521	0	0	0
Unique scores	Median	4,521	0	0	0
Unique scores	Geometric	4,521	0	0	0
Unique scores	Borda	4,521	11	+0.0024	1
Unique fullnames	Mean	4,496	0	0	0
Unique fullnames	Median	4,496	0	0	0
Unique fullnames	Geometric	4,496	0	0	0
Unique fullnames	Borda	4,496	7	-0.0002	1

Table 5: Monte Carlo precision of HELM random-order mean stopping costs, in affected subjects. All 79 fixed targets, including initial winners, are averaged; each uses 2,000 orders (seed 519733), shared across rules within target. SEs and approximate marginal normal 95% intervals quantify permutation Monte Carlo error conditional on these scores and targets. They do not describe model-population or future-task uncertainty, simultaneous coverage, or between-rule differences. Open LLM expectations are exact and are not included.

Rule	Mean cost	MC SE	Marginal 95% interval
Mean	27.2314	0.0052	[27.2213, 27.2415]
Median	21.8963	0.0037	[21.8890, 21.9036]
Geometric	29.9356	0.0063	[29.9233, 29.9480]
Borda	47.8901	0.0034	[47.8834, 47.8968]

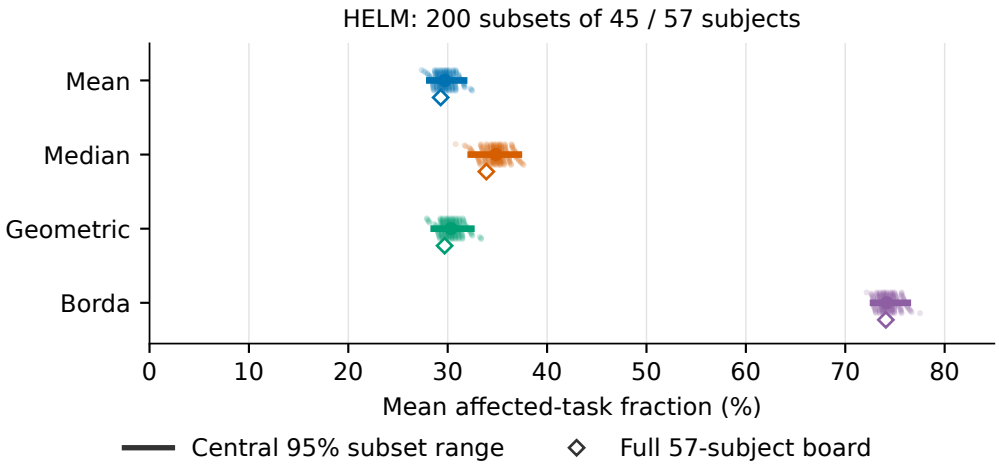


Figure 4: Task-selection sensitivity on HELM under perfect-score saturation. For each of 200 uniform subsets of 45 distinct subjects, all 79 targets are re-optimized. Faint dots are the across-target mean affected-task fraction for each subset; thick segments span the 2.5th–97.5th empirical percentiles and solid dots mark medians. Open diamonds show the full 57-subject board, offset vertically for visibility. These are perturbation ranges over the observed subjects, not confidence intervals for future tasks, item-level accuracy, or independent models. Seed: 449027.

C Reproduction and provenance

The delivery includes the complete analysis code, processed data, per-target results, exact-solver tests and provenance under `code/`; publication generators and their frozen numerical inputs are under `presentation/`. Root `README.md` gives the build and reproduction commands. Every included figure and table is generated by `presentation/render.py`; `presentation/README.md` maps each asset to its numerical source, and `presentation/source_manifest.json` records upstream revision and hashes. Captions are provided in `presentation/captions.tex`.

The analysis manifest pins HELM v1.13.0 and the full Open LLM snapshot commit, and verifies raw download checksums before processing. Only public score summaries are downloaded; raw per-model inference is not repeated. The processed matrices include all complete HELM entries and the unflagged Open LLM entries, with no imputation. The separate Open LLM `results` snapshot is available in the task source map but is not used: the six aggregate tasks in `contents` define this study's intervention unit.

Environment versions, measured runtime records and fixed random seeds accompany the code. Random task-order seed is 519733; subject-subset seed is 449027; exhaustive diagnostic seeds are 7821, 999 and 112. Primary optima are deterministic, though different solver versions can select different equally optimal witnesses. No failed numerical run, unfinished fit, or model-training experiment is represented as completed evidence. Before execution, the subset diagnostic was changed from sampling with replacement to distinct-subject omission because duplicated subjects should not become independently attackable coordinates.

The numerical jobs were allocated at most two logical CPUs and 4 GiB memory per worker on a shared Linux x86-64 host, with no GPU. BLAS/OpenMP thread limits were one. A later worker probe reports an Intel Xeon Gold 6230 at 2.10 GHz; individual historical job CPU model strings were not retained. Allocation is not measured peak usage: per-job peak RSS and project-wide CPU-hours were not recorded. The raw inputs comprise 122 files totaling 37,800,558 bytes. Full configuration and timing scopes are in `code/notes/COMPUTE_AND_REPRODUCTION.md` and `code/results/compute_configuration.json`.

Recorded component times were 12.82 s for the HELM primary/sensitivity bundle, 359.13 s for the Open LLM bundle (including alternate representations), 149.30 s for subject subsets, and 1.04 s/14.03 s for HELM/Open LLM random baselines. Their sum is 536.32 s, not measured end-to-end elapsed time. The two pilots and a relocated test run add about three seconds; acquisition, preparation, smaller audits and installation were not individually timed in the original study. As a conservative planning estimate on a comparable worker, allow 20 minutes for the scientific pipeline with dependencies installed, or 30–60 minutes including a new environment and downloads; network contention can exceed this estimate. Figure rendering and the PDF build are separate steps. These estimates describe reproduction, not model training.

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D Responsible interpretation and research-process disclosure

The intended benefit is a transparent audit of how aggregation and improvement assumptions interact. The released optimizer could also help a developer identify favorable task subsets for leaderboard gaming. We therefore present it as conditional analysis, expose ceiling and feasibility dependence, and make no allegations about individual models' training histories. Overinterpreting a low conditional cost as evidence of contamination could unjustly damage developers; it is unsupported by these data. Evaluation designers should combine aggregate-rule auditing with independent checks of training provenance, broader task coverage and explicit reporting policies. We release no new trained model, private personal dataset or evaluation answers.

The public aggregate-score snapshots are used for reproducible scientific audit and critical assessment. A separate license for those score datasets was not established; our code/data-access instructions do not assert that public accessibility grants unrestricted reuse. No human participants were recruited. The NeurIPS Code of Ethics was read on 3 November 2026 at `https://neurips.cc/public/EthicsGuidelines`; the checklist retains the qualification about unestablished data licensing rather than certifying blanket compliance.

Autonomous LLM agents were involved in interpreting the objective, designing the conditional model and comparisons, writing and checking code, analyzing results, constructing figures, drafting the manuscript and conducting internal reviews. This goes beyond language editing. Reported empirical values come from executed deterministic or explicitly randomized code and retained public-score inputs, not generated numerical assertions. The agents' contributions do not make them human authors or establish independent human validation; the disclosed review process and reproducible artifacts are the means for scrutiny.

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